Open problems:
Unphysicality of Hilbert spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Further problems.** The list of ways in which QSTO:H fails the base conditions is open-ended: "If the reader is aware of other problems that should be added, do contact us." (Brief, Sec. 1, footnote 1.)

2. **A physically adequate model.** The brief concludes that "the solution is not to find workarounds for these issues, but to fix the mathematical model to better reflect the physical requirements". A replacement for QSTO:H that satisfies [B]COBS, [B]WDQ and [B]STDIST is not given here; one candidate is the brief "Physical vector spaces for quantum mechanics". (Brief, Sec. 5.)

3. **Expectations of a general unbounded observable.** [B]COBS is about the expectation $\psi \mapsto \langle\psi|O|\psi\rangle$, while "boundedness is equivalent to continuity" is about everywhere-defined linear maps; an unbounded observable O is only densely defined. That the expectation of every unbounded observable is discontinuous is not proved. The formalization covers everywhere-defined linear maps (`not_continuous_of_unbounded`) and the oscillator example for N only. (Brief, Sec. 3, proof of Theorem 1, first paragraph; Log, "Results".)

From the formalization

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **The conditions themselves.** [B]COBS, [B]WDQ, [B]STDIST and the inconsistency claimed in Theorem 1 are informal; only the mathematics behind each of the three arguments is checked. (Log, "Setting".)

2. **The unitary between the two frames.** The unitary U that implements the frame change $y(x) = \tan(\frac{\pi}{2}\operatorname{erf}(x))$ on wave functions, and the change of variables in the integral, are not formalized; only the densities $|\psi|^2$, $|\phi|^2$ and eq. (2) are. So is the general claim that "we can find a unitary U that connects operators with different domains". (Brief, Sec. 3; Log, "Differences".)

3. **Point-wise values.** That $\psi(x)$, and so $|\psi(x)|^2$, is not defined at each point of a ray of L^2 is not formalized. (Brief, Sec. 4; Log, "Differences", last item.)

4. **Commutator relations.** That $[X, P]/(i\hbar) = I$ cannot hold on the whole of $\mathcal{H}$ and must be restricted to a domain is not formalized. (Brief, Sec. 4; Log, "Differences", last item.)

5. **Domain of the Schrödinger equation.** That the Schrödinger equation holds only on the domain of H, since $P\psi$ and $V(x)\psi(x)$ need not be defined, is not formalized. (Brief, Sec. 4; Log, "Differences", last item.)